

CS 361 - Homework 01 - Theory

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1 Overview

This homework consists of a few theory questions. The focus is on possible scenarios of execution for atomic statements.

All answers must be **typed** and submitted as a single **PDF**. The PDF must include your name and all questions must be clearly labeled. You do not have to repeat the question text, only label the questions clearly.

For all algorithm examples, lines that are not given numbers are considered starting conditions and not part of the algorithm. For example, if an algorithm has a local variable that starts at value $x = 10$, this might not be given a line number. It is a started condition for the threads. In practice, it would be done before the threads starts.

All lines are assumed to be **atomic statements**. Execution will never context switch in the middle of completing a numbered line.

2 Questions

This homework is worth 50 points.

Question 1 : 20 points

Use the following algorithm to answer this question. The statement **for i from a to b** means i iterates through all values in the range including a and b . In python this would be **for i in range(a,b+1)** and in C it would be **for(int i=a; i <= b; i++;)**. The loop exits when $i = b + 1$.

Algorithm: Concurrent Counting	
Setup: $n = 0$	
Thread P	Thread Q
local variable temp, i	local variable temp, j
p1: for i from 1 to 10 do	q1: for j from 1 to 10 do
p2: temp=n	q2: temp=n
p3: n=temp+1	q3: n=temp+1

Note: If your scenario table has extended repetitions you may write a row stating something like "Previous 3 lines repeat 4 times" instead of duplicating the same sequence.

- (a) (10 points) Give a Scenario where the final value of n is 10. Show the Scenario Table.
- (b) (10 points) Give a Scenario where the final value of n is 2. Show the Scenario Table.

Scenario for a					
P	Q	Instruction	n	temp p	temp q
p1	q1	Initialization	0	-	-
p2	q1	p1: for i from 1 to 10 do	0	-	-
p2	q2	q1: for i from 1 to 10 do	0	-	-
p3	q2	p2: temp=n	0	0	-
p3	q3	q2: temp=n	0	0	0
p1	q3	p3: n=temp+1	1	0	0
p1	q1	q3: n=temp+1	1	0	0
Previous 6 lines repeat 10 times					
-	-	exited	10	9	9

Scenario for b					
P	Q	Instruction	n	temp p	temp q
p1	q1	Initialization	0	-	-
p2	q1	p1: for i from 1 to 10 do	0	-	-
p3	q1	p2: temp=n	0	0	-
p3	q2	q1: for i from 1 to 10 do	0	0	-
p3	q3	q2: temp=n	0	0	0
p3	q1	q3: n=temp+1	1	0	0
Do q1 to q3 8 more times					
p1	q1	p3: n=temp+1	1	0	9
p1	q2	q1: for i from 10 to 10 do	1	0	9
p1	q3	q2: temp=n	1	0	1
p2	q1	p1: for i from 2 to 10 do	1	0	1
p3	q2	p2: temp=n	1	1	1
p1	q1	p3: n=temp+1	2	1	1
Do p1-p3 8 more times					
p1	q1	q3: n=temp+1	2	9	1
p2	q1	p1:for i from 11 to 10 do	2	9	1
-	q1	exit	2	9	1
-	q2	q1:for i from 11 to 10 do	2	9	1
-	-	exit	2	9	1

Question 2 : 30 points

Three algorithms that all solve the same problem are given below.

Assume that for the function f , there is one unique integer value $i \in \mathbb{Z}$ for which $f(i) = 0$. Each of the following algorithms searches for i .

Each of the algorithms does not work as intended.

Provide one counterexample to show why the algorithm does not work correctly. To work correctly, **both** threads **must** exit and it **must** be obvious what the correct value of i is.

State any assumptions before the start of the scenario table. For example, you might say something like "assume Thread P has found $f(i)=0$ to be true on line p4." Then start the scenario table on line p4. Your scenario table should focus on showing the flaw of the algorithm happening.

Note: The instruction **await turn==2 then turn=1** is a single atomic operation that busy waits. It loops on that line if the condition is false. If the condition is true, it execute the assignment and moves on. It is a single line version of the following code:

```

1 while(turn != 2){
2     //Just wait here!
```

```

3 }
4 turn = 1

```

(a) (10 points) Write a Scenario Table showing this algorithm does not work.

Algorithm Zero A	
Setup: global boolean found	
Thread P	Thread Q
local variable i=0 p1: found = false p2: while not found p3: i=i+1 p4: found = (f(i)==0)	local variable j=1 q1: found = false q2: while not found q3: j=j-1 q4: found = (f(j)==0)

Scenario for Algorithm Zero A					
P	Q	Instruction	i	j	found
p1	q1	Initialization	0	1	-
p1	q2	q1: found = false	0	1	false
p1	q3	q2: while not found	0	1	false
p1	q4	q3: j=j-1	0	0	false
p1	q2	q4: found = (f(j)==0)	0	0	true
p2	q2	p1: found = false	0	0	false
Assuming that the desired result is i = 0 this algorithm will run forever					

(b) (10 points) Write a Scenario Table showing this algorithm does not work.

Algorithm Zero B	
Setup: boolean found=false	
Thread P	Thread Q
local variable i=0 p1: while not found p2: i=i+1 p3: found = (f(i)==0)	local variable j=1 q1: while not found q2: j=j-1 q3: found = (f(j)==0)

Assuming that i = 0 is the number we need to find

Scenario for Algorithm Zero B					
P	Q	Instruction	i	j	found
p1	q1	Initialization	0	1	false
p1	q2	q1: while not found	0	1	false
p1	q3	q2: j=j-1	0	0	false
p2	q3	p1: while not found	0	0	false
p3	q3	p2: i=i+1	1	0	false
p3	q1	q3: found = (f(j)==0)	1	0	true
p1	q1	p3: found = (f(i)==0)	1	0	false
p1	q2	q1:while not found	1	0	false
this algorithm will run forever					

(c) (10 points) Write a Scenario Table showing this algorithm does not work.

Algorithm Zero C	
Setup: boolean found=false, turn=1	
Thread P	Thread Q
local variable i=0	local variable j=1
p1: while not found	q1: while not found
p2: await turn==1	q2: await turn==2
then turn=2	then turn=1
p3: i=i+1	q3: j=j-1
p4: if f(i)==0	q4: if f(j)==0
p5: found = true	q5: found = true

Assuming that the number we need is 1

Scenario for Algorithm Zero C						
P	Q	Instruction	i	j	found	turn
p1	q1	Initialization	0	1	false	1
p2	q1	p1: while not found	0	1	false	1
p3	q1	p2: await turn==1, then turn=2	0	1	false	2
p4	q1	p3: i = i+1	1	1	false	2
p5	q1	p4: if f(i)==0	1	1	false	2
p5	q2	q1: while not found	1	1	false	2
p5	q3	q2: await turn==2, then turn=1	1	1	false	1
p5	q4	q3: j = j-1	1	0	false	1
p5	q5	q4: if f(j)==0	1	0	false	1
skip q5 since f(0) != 0						
p5	q2	q1: while not found	1	0	false	1
p1	q2	p5: found = true	1	0	true	1
p2	q2	p1: while not found	1	0	true	1
-	q2	exited	1	0	true	1
-	q3	q2: await turn==2, then turn=1	1	0	true	1
This algorithm will run forever						